



Agent Prompt: Organization, Inventory Activation, Billing, and Branches

Copy and provide the following prompt to your implementation agent.

Important Override Instruction

You are modifying the existing Orvionhub codebase.

The current authentication and general onboarding system already exists and should remain intact. Users can:

- Register.
- Log in.
- Verify email.
- Manage their personal profile.
- Manage sessions and security.
- Reset passwords.
- Access the central Orvionhub account system.

Do not rebuild or replace the existing user authentication and personal onboarding system.

However, the existing organization, application activation, billing, or branch implementation may be incomplete, inconsistent, or based on an older design.

For this task, the implementation below is authoritative and must override conflicting existing organization, application, billing, branch, and onboarding behavior.

Do not preserve conflicting behavior merely because it already exists.

1. Authoritative Product Structure

Orvionhub must use this hierarchy:

```
User account
├── Organization / Workspace
│   ├── Billing account
│   │   ├── Subscription
│   │   │   ├── Plan entitlements
│   │   │   └── Application access
```

Definitions:

- A **user** is a personal Orviohub account.
- An **organization** and **workspace** represent the same business container.
- A **billing account** belongs to the organization.
- A **subscription** controls the organization's commercial plan.
- An **application** belongs to an organization.
- A **branch** belongs to an application within an organization.
- A **team member** belongs to the organization, application, and selected branches.
- Personal account data must remain separate from organization data.

For the current MVP:

Available application:

- Inventory

Unavailable applications:

- Task Management
- CRM
- Gym Management
- Booking
- Other future applications

Regular users must see and access only Inventory. Future applications may remain in the internal registry but must be hidden and unavailable.

The PRD defines the MVP around products, stock, purchases, sales, customers, suppliers, branches, reports, and basic business operations—not a complete ERP.^[1]

2. Existing Authentication Boundary

Keep the existing central account system.

Do not modify the existing behavior for:

- User registration.
- Login.
- Logout.
- Email verification.
- Password reset.
- Personal profile.
- Personal sessions.

- Personal security.
- Global account deletion.
- Connected login methods.

Organization creation begins only after the user has an authenticated Orviohub account.

Required account state before organization creation:

```
authenticated
emailVerified
accountStatus = active
```

Phone verification is not required for organization creation in the MVP.

3. Organization Creation Rules

A user can create an organization only when the backend confirms they are eligible.

The ownership limit applies to organizations owned or billed by the user—not organizations where they are merely a member.

Current MVP rules:

```
Maximum owned organizations: 3
Maximum Free Trial organizations: 1
Joined organizations: unlimited, subject to the organization's own rules
```

A joined organization must not consume the user's ownership limit.

Organization creation flow

```
Authenticated user
↓
Open workspace selector
↓
Select "Create organization"
↓
Backend checks ownership eligibility
↓
User enters organization details
↓
User selects Free Trial or Standard
↓
Backend creates organization
↓
Billing context is created
↓
Inventory application is prepared
↓
Primary branch is created
```

```
↓
Owner membership is created
↓
Inventory onboarding starts
```

Required organization fields

```
name
slug
type
category
country
state
city
timezone
currency
businessPhone optional
businessEmail optional
address optional
logo optional
```

Recommended defaults:

```
type = "business"
country = "NG"
currency = "NGN"
timezone = "Africa/Lagos"
status = "creating"
```

Organization creation must create

1. Organization record.
2. Owner membership.
3. Default organization settings.
4. Default roles.
5. Billing account.
6. Subscription context.
7. Inventory application record.
8. Primary Inventory branch.
9. Inventory onboarding flow.
10. Audit event.
11. Welcome notification.

The operation must be idempotent. Retrying the same request must not create duplicate organizations, subscriptions, branches, or owner memberships.

4. Organization Status

Use these organization statuses:

```
creating
setup_incomplete
trial
active
past_due
suspended
archived
deleting
deleted
```

Meaning:

- `creating`: initial provisioning is running.
- `setup_incomplete`: organization exists but onboarding is incomplete.
- `trial`: organization is using Free Trial entitlements.
- `active`: organization has an active plan.
- `past_due`: payment failed but grace period is active.
- `suspended`: access is restricted.
- `archived`: organization is retained but inactive.
- `deleting`: deletion workflow is in progress.
- `deleted`: organization is no longer available.

Do not use only a frontend flag to determine organization status.

5. Plans and Billing

Billing belongs to the organization or billing account.

Do not implement the old model where one user subscription independently controls all organizations.

Correct structure:

```
Organization
├── Billing account
│   └── Subscription
│       └── Plan
│           └── Entitlements
```

Employees and invited team members do not purchase separate subscriptions.

Current plans

Free Trial

Duration: 30 days
Owned organizations: 1 allowed for trial eligibility
Applications: Inventory only
Branches: 1
Members: 2
Products: 500
Monthly transactions: 300

Standard

Price: ₦7,500 monthly
Price: ₦75,000 annually
Owned organizations: up to 3
Applications: up to 3 per organization, subject to availability
Branches: up to 3 per organization
Members: 10 per organization
Products: 5,000 per organization
Monthly transactions: 5,000 per organization

For the current MVP, only Inventory is available, even if the Standard plan architecture supports more applications later.

Store money in the smallest currency unit:

₦7,500 = 750000 kobo

Do not collect card details in Orviohub. Paystack handles payment details.

6. Free Trial Journey

User journey

```
graph TD
    A[User signs in] --> B[Creates organization]
    B --> C[Selects Free Trial]
    C --> D[Backend checks one-Free-Trial eligibility]
    D --> E[Organization is created]
    E --> F[Subscription status = trialing]
```

```
Active plan = free_trial
↓
Inventory entitlement is created
↓
Primary Inventory branch is created
↓
Inventory onboarding starts
↓
User enters Inventory dashboard
```

Expected trial state

```
selectedPlan: free_trial
checkoutStatus: not_required
paymentStatus: not_required
subscriptionStatus: trialing
activePlan: free_trial
entitlementStatus: active
trialStart: timestamp
trialEnd: trialStart + 30 days
```

No Paystack checkout is required for Free Trial.

Trial restrictions

When the trial ends:

- Do not delete the organization.
- Do not delete products.
- Do not delete branches.
- Preserve historical records.
- Restrict creation of new resources.
- Display a clear upgrade prompt.
- Allow the owner to upgrade to Standard.
- Use read-only or limited operational access according to the billing policy.

7. Free Trial to Standard Upgrade

User journey

```
User opens organization billing
↓
Selects "Upgrade to Standard"
↓
Selects monthly or annual billing
↓
Backend creates pending checkout context
```

```
↓
Backend initializes Paystack
↓
Paystack checkout opens
↓
User completes payment
↓
Paystack returns reference
↓
Backend verifies payment server-side
↓
Webhook is received and verified
↓
Billing event is deduplicated
↓
Subscription becomes active
↓
Standard entitlements are created
↓
Organization becomes active
↓
Inventory limits are updated
↓
User returns to Inventory
```

Expected post-payment state

```
selectedPlan: standard
checkoutStatus: completed
paymentStatus: success
subscriptionStatus: active
activePlan: standard
entitlementStatus: active
trialEnd: null
```

The user dashboard and superadmin dashboard must read this same authoritative billing state.

Never activate Standard access based only on the frontend Paystack callback.

Required payment verification

The backend must verify:

- Paystack reference.
- Payment status.
- Amount.
- Currency.
- Customer email.
- Billing account.
- Plan.

- Billing interval.
- Transaction uniqueness.
- Webhook signature.
- Provider event ID.
- Subscription state.

Paystack payment must be verified server-side, and webhook events must be stored and deduplicated.^[2]

8. Billing State Model

Use explicit states.

Checkout status

```
not_required
not_started
initialized
redirected
completed
failed
expired
```

Payment status

```
not_required
pending
success
failed
refunded
under_review
```

Subscription status

```
trialing
active
past_due
grace_period
cancelled
expired
suspended
```

Entitlement status

```
pending
active
restricted
expired
revoked
```

State transitions

```
trialing → active
trialing → expired

active → past_due
past_due → active
past_due → grace_period
grace_period → active
grace_period → suspended

active → cancelled
cancelled → expired
```

Do not immediately delete or destroy business data after payment failure.

9. Billing Data Model

Implement or reconcile these tables.

Billing accounts

```
billingAccounts: {
  ownerUserId: Id<"users">,
  workspaceId: Id<"workspaces">,
  billingEmail: string,
  billingPhone?: string,
  provider: "paystack",
  providerCustomerId?: string,
  currency: string,
  status: "active" | "suspended" | "closed",
  createdAt: number,
  updatedAt: number
}
```

Subscriptions

```
subscriptions: {
  billingAccountId: Id<"billingAccounts">,
  workspaceId: Id<"workspaces">,
  planId: string,
```

```
provider: "paystack",
providerSubscriptionId?: string,
status: string,
billingInterval: "monthly" | "annual",
currentPeriodStart?: number,
currentPeriodEnd?: number,
trialStart?: number,
trialEnd?: number,
gracePeriodEnd?: number,
cancelAtPeriodEnd?: boolean,
createdAt: number,
updatedAt: number
}
```

Payments

```
payments: {
  billingAccountId: Id<"billingAccounts">,
  workspaceId: Id<"workspaces">,
  subscriptionId: Id<"subscriptions">,
  provider: "paystack",
  providerReference: string,
  amount: number,
  currency: string,
  status: "pending" | "success" | "failed" | "refunded",
  paidAt?: number,
  metadata?: any,
  createdAt: number,
  updatedAt: number
}
```

Billing events

```
billingEvents: {
  provider: "paystack",
  providerEventId: string,
  eventType: string,
  workspaceId?: Id<"workspaces">,
  billingAccountId?: Id<"billingAccounts">,
  subscriptionId?: Id<"subscriptions">,
  status: "received" | "processed" | "ignored" | "failed",
  payloadMetadata?: any,
  processedAt?: number,
  errorMessage?: string,
  createdAt: number
}
```

Make providerEventId unique to prevent duplicate processing.

Entitlements

```
workspaceEntitlements: {  
  workspaceId: Id<"workspaces">,  
  planId: string,  
  featureKey: string,  
  limitValue?: number,  
  limitType: "fixed" | "unlimited" | "boolean",  
  enabled: boolean,  
  status: "active" | "expired" | "revoked",  
  effectiveFrom: number,  
  effectiveUntil?: number,  
  createdAt: number,  
  updatedAt: number  
}
```

Do not scatter checks such as:

```
if (plan === "standard") {  
  // allow  
}
```

Use a centralized entitlement service.

10. Inventory Application Activation

Inventory is the only application available to normal users.

Inventory activation flow

```
Authenticated user  
  ↓  
Active workspace membership  
  ↓  
Workspace status allows access  
  ↓  
Billing status allows access  
  ↓  
Inventory entitlement exists  
  ↓  
Inventory application is available  
  ↓  
Inventory record is created or activated  
  ↓  
Inventory onboarding starts
```

Inventory application record

```
workspaceProducts: {  
  workspaceId: Id<"workspaces">,  
  productKey: "inventory",  
  status: "activating" | "setup_incomplete" | "active" | "suspended" | "deactivated"  
  activatedBy: Id<"users">,  
  activatedAt?: number,  
  suspendedAt?: number,  
  createdAt: number,  
  updatedAt: number  
}
```

Do not create active records for Task Management, CRM, Gym, Booking, or other future applications.

Inventory access requires

```
authenticated user  
+ active workspace membership  
+ active Inventory application access  
+ valid organization status  
+ valid billing entitlement  
+ branch access where required
```

11. Branch Creation

Branches belong to the organization and Inventory application.

Use this relationship:

```
Organization  
  └─ Inventory  
      └─ Branch
```

Primary branch

The first Inventory activation must create one primary branch.

```
branch.status = active  
branch.isPrimary = true
```

The user may rename the primary branch during onboarding.

Branch creation journey

```
User opens Inventory settings
↓
Selects "Add branch"
↓
Backend checks Standard or eligible entitlement
↓
User enters branch details
↓
Backend validates branch limit
↓
Branch is created
↓
Branch settings are created
↓
Audit event is written
↓
User may assign staff
```

Branch fields

```
workspaceId
productKey = "inventory"
name
code
address
phone
email
managerId optional
isPrimary
status
createdAt
updatedAt
```

Branch rules

- Free Trial allows one branch.
- Standard allows up to three branches.
- A user cannot create a branch in another organization.
- A branch must belong to an active Inventory application.
- The last active primary branch cannot be archived.
- Archiving preserves historical data.
- Branch removal must not delete sales, purchases, stock movements, or audit records.
- Branch permissions are separate from workspace membership.

Branch access

A user may be:

```
workspace member
+ Inventory member
+ member of selected branches
```

Being a workspace member does not automatically give Inventory access.

Being an Inventory member does not automatically give access to every branch.

12. Inventory Onboarding

Use this flow:

```
Inventory welcome
  ↓
Business category
  ↓
Business profile
  ↓
Primary branch setup
  ↓
Currency and pricing
  ↓
Inventory policies
  ↓
Product setup
  ↓
CSV import or manual products
  ↓
Opening stock
  ↓
Staff invitations
  ↓
Receipt settings
  ↓
First-sale tutorial
  ↓
Completion
```

The flow must be resumable.

Store:

```
flowId
userId
workspaceId
productKey = "inventory"
currentStep
completedSteps
```

```
skippedSteps
failedSteps
flowVersion
status
startedAt
lastUpdatedAt
completedAt
```

Required steps cannot be skipped. Optional staff invitation and product setup may be skipped and completed later.

13. Team and Permissions

Organization membership

Answers:

```
Does the person belong to the organization?
```

Inventory membership

Answers:

```
Can the person use Inventory?
```

Branch membership

Answers:

```
Which branches can the person access?
```

Inventory roles

```
Inventory Owner
Inventory Manager
Sales Attendant
Stock Manager
Accountant
Inventory Viewer
```

Inventory permissions

```
inventory.view
inventory.manage_products
inventory.view_stock
inventory.receive_stock
inventory.adjust_stock
inventory.record_sales
inventory.cancel_sales
inventory.process_returns
inventory.view_cost
inventory.view_profit
inventory.view_reports
inventory.export_data
inventory.manage_members
inventory.manage_branch_access
inventory.manage_settings
```

The backend must calculate effective permissions from:

```
workspace membership
+ workspace role
+ Inventory membership
+ Inventory role
+ branch membership
+ branch role
+ explicit permissions
```

14. User Stories

Organization creation

As an authenticated Orvionhub user, I want to create an organization so that I can manage my business separately from my personal account.

As a user, I want the system to tell me whether I can create another organization before I complete the form.

As a user, I should not be able to create a fourth owned organization.

As a user, I should still be able to join an organization owned by somebody else.

As an organization owner, I should automatically become the organization owner and Inventory administrator.

As an organization owner, I want the organization creation process to create the default settings, billing context, Inventory application, and primary branch together.

Free Trial

As a new organization owner, I want to start one Free Trial without entering payment details.

As a user, I want to know when my Free Trial ends.

As a trial owner, I want to use Inventory within the Free Trial limits.

As a user, I want to upgrade before my trial expires without losing my data.

Standard upgrade

As a trial owner, I want to upgrade my organization to Standard through Paystack.

As a user, I want my Standard access to activate only after Orviohub verifies my payment.

As a user, I want the same active plan to appear in both my billing dashboard and the superadmin dashboard.

As a user, I want my existing products, branches, staff, and transactions preserved after upgrading.

As a user, I want Standard entitlements to become available immediately after successful verification.

Inventory activation

As an organization owner, I want Inventory to be the only available application in the MVP.

As a user, I want to enter Inventory after organization setup without seeing unfinished applications.

As a user, I want Inventory onboarding to resume from my last completed step.

Branches

As an Inventory owner, I want the system to create a primary branch during setup.

As an Inventory owner, I want to add branches according to my plan limit.

As a branch manager, I want to update branch contact information without changing the organization profile.

As a user, I want branch data to be isolated so that sales and stock belong to the correct location.

As an owner, I want to archive a branch without losing its historical transactions.

Superadmin

As a superadmin, I want to see every organization and its owner.

As a superadmin, I want to see the organization's active plan, payment state, entitlement state, and billing history.

As a superadmin, I want to see whether Inventory is active, incomplete, suspended, or failed.

As a superadmin, I want to see every branch and its active member count.

As a superadmin, I want to see organization, application, branch, membership, onboarding, and billing audit events.

As a superadmin, I want to suspend or restore an organization using a confirmed and audited action.

As a superadmin, I must not see passwords, raw tokens, TOTP secrets, or payment card information.

15. Data Coming Into the System

Account data

Already managed by the existing authentication system:

```
userId
email
emailVerifiedAt
name
accountStatus
```

Do not duplicate this unnecessarily.

Organization creation input

```
{
  "name": "Code X Stores",
  "type": "business",
  "category": "retail",
  "country": "NG",
  "state": "Enugu",
  "city": "Enugu",
  "timezone": "Africa/Lagos",
  "currency": "NGN",
  "businessPhone": "+234...",
  "businessEmail": "business@example.com",
  "address": "..."
```

Billing input

```
{
  "planKey": "free_trial",
  "billingInterval": null
}
```

or:

```
{
  "planKey": "standard",
  "billingInterval": "monthly"
}
```

Branch input

```
{
  "name": "Main Store",
  "code": "MAIN",
  "address": "...",
  "phone": "+234...",
  "email": "branch@example.com"
}
```

Staff invitation input

```
{
  "email": "staff@example.com",
  "productKey": "inventory",
  "role": "sales_attendant",
  "branchIds": ["branch_id"]
}
```

Every input must be:

- Validated.
- Normalized.
- Authorized.
- Workspace-scoped.
- Idempotent where retries are possible.
- Audited when it changes state.

16. Data Going Out of the System

Organization context response

```
{
  "workspace": {
    "id": "ws_123",
    "name": "Code X Stores",
    "status": "trial",
    "currency": "NGN",
    "timezone": "Africa/Lagos"
  },
  "membership": {
    "role": "owner",
    "status": "active"
  },
  "billing": {
    "plan": "free_trial",
    "subscriptionStatus": "trialing",
    "entitlementStatus": "active",
    "trialEnd": 1790000000000
  },
  "applications": [
    {
      "key": "inventory",
      "status": "setup_incomplete",
      "access": true
    }
  ],
  "branches": [
    {
      "id": "branch_123",
      "name": "Main Store",
      "isPrimary": true,
      "status": "active"
    }
  ],
  "permissions": [
    "inventory.view",
    "inventory.record_sales",
    "inventory.manage_settings"
  ]
}
```

Billing response

```
{
  "plan": {
    "key": "standard",
    "name": "Standard"
  },
  "subscription": {
    "status": "active",
  }
}
```

```

    "billingInterval": "monthly",
    "currentPeriodEnd": 1790000000000
  },
  "payment": {
    "status": "success",
    "currency": "NGN"
  },
  "entitlements": {
    "maxBranches": 3,
    "maxMembers": 10,
    "maxProducts": 5000,
    "maxMonthlyTransactions": 5000
  },
  "usage": {
    "branches": 2,
    "members": 5,
    "products": 200,
    "monthlyTransactions": 800
  }
}

```

Never return:

- Password hashes.
- Raw session tokens.
- Paystack secret keys.
- Webhook secrets.
- Raw payment card data.
- TOTP secrets.
- Invitation tokens.
- Private provider credentials.

17. API Endpoints

Organization

```

GET    /v1/workspaces
POST   /v1/workspaces
GET    /v1/workspaces/:workspaceId
PATCH /v1/workspaces/:workspaceId
POST   /v1/workspaces/:workspaceId/select
POST   /v1/workspaces/:workspaceId/archive
POST   /v1/workspaces/:workspaceId/restore

```

Organization eligibility

```
GET /v1/entitlements/can-create-workspace
GET /v1/entitlements/free-trial-eligibility
```

Eligibility reads should not create audit noise.

Inventory application

```
GET /v1/workspaces/:workspaceId/applications
GET /v1/workspaces/:workspaceId/applications/inventory
POST /v1/workspaces/:workspaceId/applications/inventory/activate
POST /v1/workspaces/:workspaceId/applications/inventory/suspend
POST /v1/workspaces/:workspaceId/applications/inventory/restore
```

Do not expose future application activation endpoints to regular users.

Branches

```
GET /v1/workspaces/:workspaceId/branches
POST /v1/workspaces/:workspaceId/branches
GET /v1/workspaces/:workspaceId/branches/:branchId
PATCH /v1/workspaces/:workspaceId/branches/:branchId
POST /v1/workspaces/:workspaceId/branches/:branchId/set-primary
POST /v1/workspaces/:workspaceId/branches/:branchId/suspend
POST /v1/workspaces/:workspaceId/branches/:branchId/restore
POST /v1/workspaces/:workspaceId/branches/:branchId/archive
```

Billing

```
GET /v1/workspaces/:workspaceId/billing
GET /v1/workspaces/:workspaceId/billing/usage
GET /v1/workspaces/:workspaceId/billing/invoices
POST /v1/workspaces/:workspaceId/billing/checkout
POST /v1/workspaces/:workspaceId/billing/verify
POST /v1/billing/webhooks/paystack
POST /v1/workspaces/:workspaceId/billing/change-plan
POST /v1/workspaces/:workspaceId/billing/cancel
POST /v1/workspaces/:workspaceId/billing/resume
```

Onboarding

```
GET /v1/workspaces/:workspaceId/onboarding/inventory
POST /v1/workspaces/:workspaceId/onboarding/inventory/start
PATCH /v1/workspaces/:workspaceId/onboarding/inventory/progress
POST /v1/workspaces/:workspaceId/onboarding/inventory/complete-step
POST /v1/workspaces/:workspaceId/onboarding/inventory/skip-step
POST /v1/workspaces/:workspaceId/onboarding/inventory/complete
```

Members and invitations

```
GET    /v1/workspaces/:workspaceId/members
POST   /v1/workspaces/:workspaceId/invitations
GET    /v1/workspaces/:workspaceId/invitations
POST   /v1/workspaces/:workspaceId/invitations/:id/resend
POST   /v1/workspaces/:workspaceId/invitations/:id/revoke
PATCH /v1/workspaces/:workspaceId/members/:membershipId
POST   /v1/workspaces/:workspaceId/members/:membershipId/suspend
POST   /v1/workspaces/:workspaceId/members/:membershipId/restore
DELETE /v1/workspaces/:workspaceId/members/:membershipId
```

18. Superadmin Dashboard Requirements

Superadmin must be able to inspect every organization.

Organization list

Display:

- Organization name.
- Organization ID.
- Owner.
- Owner email.
- Organization status.
- Plan.
- Subscription status.
- Payment status.
- Trial end.
- Active applications.
- Branch count.
- Member count.
- Onboarding status.
- Last activity.
- Creation date.

Organization detail

Display:

Overview

- Organization profile.
- Owner.
- Status.
- Created date.
- Last activity.
- Warnings.
- Support notes.

Applications

- Inventory status.
- Activation date.
- Onboarding status.
- Application members.
- Application permissions.
- Application errors.
- Application usage.

Future applications may appear to superadmin in the registry but must remain hidden from regular users.

Branches

For every branch:

- Branch name.
- Branch ID.
- Organization.
- Application.
- Primary status.
- Branch status.
- Address.
- Phone.
- Manager.
- Active team count.
- Products.
- Transactions.
- Last activity.

Teams

Show:

- Organization members.
- Inventory members.
- Branch members.
- Roles.
- Permissions.
- Invitation status.
- Suspension status.
- Last activity.

Billing

Show:

- Selected plan.
- Active plan.
- Checkout status.
- Payment status.
- Subscription status.
- Entitlement status.
- Trial dates.
- Billing interval.
- Amount.
- Currency.
- Paystack reference.
- Payment history.
- Subscription history.
- Entitlement history.
- Billing mismatches.

Onboarding

Show:

- Current step.
- Completed steps.
- Skipped steps.

- Failed steps.
- Flow version.
- Start date.
- Last update.
- Completion date.
- Product import status.
- Opening stock status.
- Staff invitation status.
- First sale status.

Audit logs

Show:

- Actor.
- Event type.
- Entity type.
- Entity ID.
- Application.
- Branch.
- Timestamp.
- Severity.
- Request ID.
- IP address, masked where appropriate.
- User agent.
- Safe metadata.
- Reason for administrative action.

Audit logs must be append-only and must not contain secrets.

Superadmin actions

Allowed with proper permission and confirmation:

- View organization.
- View application.
- View branch.
- View members.
- View billing.
- View onboarding.

- Add support note.
- Suspend organization.
- Restore organization.
- Archive organization.
- Suspend Inventory.
- Restore Inventory.
- Suspend branch access.
- Restore branch access.
- Recalculate entitlements.
- Retry failed provisioning.
- Extend trial with a reason.
- Revoke organization sessions.
- Export permitted organization data.

Not allowed in the MVP:

- View passwords.
- Set passwords.
- View TOTP secrets.
- Disable 2FA directly.
- View raw tokens.
- Directly edit stock balances.
- Directly edit posted sales.
- Directly edit posted purchases.
- Silently change roles.
- Silently delete organization data.
- Impersonate users.
- Delete payment records.
- Delete audit logs.

19. Security and Confirmation

Every state-changing operation must follow:

```
Authenticate admin
  ↓
Verify admin permission
  ↓
Resolve exact target
```

```
↓
Display confirmation
↓
Collect reason
↓
Require re-authentication for sensitive action
↓
Require TOTP for high-risk action
↓
Execute backend mutation
↓
Write audit event
↓
Send notification if required
```

Use:

- Normal confirmation for low-risk updates.
- Password re-authentication for sensitive changes.
- TOTP for high-risk actions.
- Confirmation phrase for destructive actions.
- Optional second-admin approval later.

Never trust the frontend confirmation alone.

20. Audit Events

Organization

```
workspace.creation_started
workspace.created
workspace.updated
workspace.archived
workspace.restored
workspace.suspended
workspace.deletion_requested
workspace.owner_assigned
workspace.entitlements_recalculated
```

Billing

```
billing.plan_selected
billing.checkout_started
billing.payment_initialized
billing.payment_verified
billing.payment_failed
billing.subscription_created
billing.subscription_renewed
billing.subscription_cancelled
```

```
billing.subscription_resumed
billing.plan_changed
billing.grace_period_started
billing.workspace_restricted
billing.entitlement_granted
billing.entitlement_revoked
billing.limit_reached
billing.webhook_received
billing.webhook_duplicate
billing.webhook_failed
```

Inventory

```
inventory.activation_started
inventory.activated
inventory.suspended
inventory.restored
inventory.onboarding_started
inventory.onboarding_completed
inventory.provisioning_failed
```

Branches

```
branch.created
branch.updated
branch.set_primary
branch.suspended
branch.restored
branch.archived
```

Memberships

```
workspace.member_invited
workspace.invitation_accepted
workspace.invitation_revoked
workspace.member_role_changed
workspace.member_suspended
workspace.member_restored
workspace.member_removed
inventory.member_added
inventory.branch_access_granted
inventory.branch_access_revoked
```

Do not audit every eligibility read. Audit actual state changes and sensitive views.

21. Required Frontend Journeys

Journey A: New Free Trial owner

Login

- Create organization
- Enter business details
- Select Free Trial
- Organization created
- Inventory activated
- Primary branch created
- Inventory onboarding
- Add products
- Add opening stock
- Optional staff invitation
- Inventory dashboard

Journey B: Existing user creates another organization

Open workspace selector

- Select Create organization
- Backend checks owned organization limit
- Enter organization details
- Select Standard
- Create organization as pending billing
- Paystack checkout
- Server verifies payment
- Standard activated
- Inventory activation
- Branch creation
- Inventory onboarding

Journey C: Free Trial upgrades

Open billing

- Select Upgrade to Standard
- Select monthly or annual
- Paystack checkout
- Payment verification
- Subscription active
- Entitlements recalculated
- Trial removed
- Standard limits shown
- Continue using Inventory

Journey D: Free Trial limit reached

```
User selects Add branch
→ Backend checks entitlement
→ Limit reached
→ Show current usage and plan limit
→ Show Standard upgrade option
→ User may upgrade
→ Existing data remains available
```

Journey E: Standard creates a branch

```
Open Inventory branch settings
→ Add branch
→ Backend checks branch entitlement
→ Enter branch details
→ Create branch
→ Create branch settings
→ Assign staff
→ Write audit event
```

Journey F: Payment fails

```
Paystack payment fails
→ Subscription becomes past_due
→ Owner receives notification
→ Grace period starts
→ New resource creation becomes restricted after policy threshold
→ Existing data remains preserved
→ User retries payment
→ If successful, subscription becomes active
```

22. Do's

- Do preserve the existing authentication system.
- Do treat the implementation above as authoritative for organization features.
- Do use workspace-based billing.
- Do create one billing account per organization for the MVP.
- Do create Inventory as the only available application.
- Do create one primary Inventory branch during activation.
- Do enforce limits server-side.
- Do use centralized entitlement checks.
- Do verify Paystack payments server-side.
- Do deduplicate Paystack webhooks.

- Do preserve data after payment failure.
- Do allow users to upgrade from Free Trial to Standard.
- Do keep joined organizations separate from owned organizations.
- Do make organization creation idempotent.
- Do use workspace-scoped queries and indexes.
- Do separate organization membership, Inventory membership, and branch membership.
- Do audit all state changes.
- Do give superadmin broad visibility without exposing secrets.
- Do use confirmations and re-authentication for dangerous actions.
- Do preserve historical inventory and financial records.
- Do use clear usage limits and upgrade prompts.

23. Don'ts

- Do not rebuild user authentication.
- Do not use the old per-user subscription model.
- Do not activate Standard based only on the frontend callback.
- Do not trust client-supplied workspace IDs, roles, permissions, or application keys.
- Do not allow future applications to appear to normal users.
- Do not create active future application memberships.
- Do not count joined organizations toward ownership limits.
- Do not create duplicate organizations on retries.
- Do not create duplicate payments from duplicate webhooks.
- Do not delete data when a trial expires.
- Do not delete data when payment fails.
- Do not silently edit posted sales, purchases, or stock.
- Do not let a workspace member access every Inventory branch automatically.
- Do not let Inventory access grant access to future applications.
- Do not allow a branch outside the current organization.
- Do not expose passwords, raw tokens, TOTP secrets, card data, or provider secrets.
- Do not let superadmin directly modify business records without a proper correction workflow.
- Do not use frontend-only billing or entitlement logic.
- Do not build unnecessary microservices for this MVP.

24. Implementation Order

Phase 1: Audit and migration

- Inspect existing organization code.
- Identify conflicting schemas and services.
- Identify per-user billing assumptions.
- Identify generic multi-application behavior.
- Create a migration plan.
- Do not delete existing data without mapping it.

Phase 2: Organization foundation

- Implement organization schema.
- Implement ownership eligibility.
- Implement organization creation mutation.
- Implement idempotency.
- Implement owner membership.
- Implement default roles.
- Implement organization settings.
- Implement audit events.

Phase 3: Billing foundation

- Implement billing account.
- Implement plans.
- Implement subscriptions.
- Implement entitlements.
- Implement Free Trial.
- Implement Standard pending billing.
- Implement centralized entitlement service.

Phase 4: Paystack

- Implement checkout initialization.
- Implement server-side verification.
- Implement webhook verification.
- Implement webhook deduplication.
- Implement payment records.

- Implement subscription state transitions.
- Implement payment failure and grace period.

Phase 5: Inventory activation

- Restrict user-facing applications to Inventory.
- Implement Inventory activation.
- Implement Inventory membership.
- Implement Inventory onboarding.
- Create primary branch.
- Add branch access checks.

Phase 6: Branches

- Implement branch creation.
- Implement branch updates.
- Implement primary branch rules.
- Implement branch limits.
- Implement branch archive and restore.
- Implement branch membership.
- Implement branch audit events.

Phase 7: Superadmin

- Organization list.
- Organization detail.
- Applications.
- Branches.
- Members.
- Billing.
- Onboarding.
- Usage.
- Audit logs.
- Support notes.
- Safe administrative actions.

Phase 8: Testing

Test:

- New Free Trial user.
- Standard user.
- Free Trial to Standard upgrade.
- Failed payment.
- Duplicate Paystack webhook.
- Third organization.
- Fourth organization rejection.
- Joined organization not consuming ownership limit.
- Inventory activation.
- Branch limit.
- Unauthorized branch access.
- Cross-organization access.
- Suspended organization.
- Archived branch.
- Removed member.
- Superadmin visibility.
- Sensitive data protection.
- Idempotent retries.

25. Final Acceptance Criteria

The implementation is complete when:

- Existing users continue using the existing authentication system.
- An authenticated user can create an organization.
- Organization creation creates owner membership, settings, billing context, Inventory access, primary branch, onboarding, and audit records.
- The first Free Trial organization can be created without payment.
- A second Free Trial organization is rejected.
- A user can own up to three organizations according to the ownership rule.
- Joined organizations do not consume ownership slots.
- Inventory is the only application visible to normal users.
- Inventory activation is separate from organization creation but integrated into onboarding.
- A primary Inventory branch is automatically created.

- Branch limits are checked server-side.
- Users can upgrade from Free Trial to Standard using Paystack.
- Standard access activates only after server-side payment verification.
- Webhooks are verified and deduplicated.
- User and superadmin billing dashboards show the same authoritative state.
- Failed payments enter the correct grace-period flow.
- Existing data is preserved during expiry, failure, suspension, and downgrade.
- Workspace, application, and branch access are separately authorized.
- Superadmin can inspect every organization, application, branch, team, billing state, onboarding state, usage state, and audit history.
- Superadmin cannot view raw authentication secrets or directly edit posted business records.
- All administrative changes are confirmed, authorized, and audited.
- Cross-organization data access is rejected.
- No conflicting old organization, application, branch, or billing behavior remains active.

This prompt should be treated as the implementation contract for the organization-to-application-to-branch foundation. The current MVP path is:

```
Existing Orviohub user
  → Create organization
  → Select Free Trial or Standard
  → Activate Inventory
  → Create primary branch
  → Complete Inventory onboarding
  → Use Inventory dashboard
```

✱✱

1. docs/Orviohub full PRD.pdf
2. docs/Orviohub Pricing Implementation Plan.pdf